

IMDADULLAH RAJI

Final-year BSc, Mechanical Engineering, BUET | expected graduation June, 2027

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Dynamical-systems and data-driven approaches to unsteady fluid flows: reduced-order modeling, active flow control, doing numerics both for generating data and analyzing them.

EDUCATION

Bangladesh University of Engineering and Technology, Dhaka

BSc in Mechanical Engineering · CGPA 3.58

2022 – 2027(est.)

Notre Dame College, Dhaka

Higher Secondary Certificate, Science

2021 year

Relevant Coursework: multivariable and vector calculus, ODEs and PDEs, complex analysis and harmonic analysis, linear algebra, numerical analysis, computer programming, engineering mechanics, fluid mechanics I&II, heat transfer, instrumentation and measurement.

RESEARCH

Modeling and Control of Flows Over Rapidly Maneuvering Airfoils

2025 – present

Undergraduate thesis · Supervisor: Dr Mohammad Ali, professor, Department of Mechanical Engineering, BUET

- Developing reduced-order models of aerodynamic forces under active actuation, with leading-edge blowing as the control input.
- Generating the training data from controlled OpenFOAM experiments at low Reynolds number, chosen so the runs stay fully resolved without turbulence modeling on available hardware.

Statistical Modeling of Lagrangian Coherent Structures

2026 – present

- Arose from the cost of computing LCS on the airfoil data above: integrating a full particle ensemble, with Jacobians and their singular values at each step, is dominated by trajectories that carry no information.
- Investigating whether early integration steps predict where attracting sets will form, allowing dense sampling to be concentrated there.

SOFTWARE AND DATA

flowkit — Python library for post-processing OpenFOAM output: snapshot loading, interpolation onto Cartesian grids, cropping, and Lagrangian analysis including trajectory integration.

OpenFOAM case files — Self-contained cases for the thesis experiments. Each documents its modeling assumptions and ships the geometry and meshing scripts needed to reproduce it.

POSITIONS

Bangladesh Olympiad on Astronomy and Astrophysics

Academic Coordinator (2026 –), Academic Member

2021 – present

- Set and evaluate problems for regional and national olympiads and IOAA team selection; run training sessions for the national team.

Team Interplanetar, BUET Mars Rover Team · Software and Automation subteam

2025

HONORS

Honorable Mention, International Olympiad on Astronomy and Astrophysics (IOAA)

2021

Bronze Medal, Open World Astronomy Olympiad (OWAO)

2022

CONFERENCE

N. A. Tanisha, **I. Raji**, A. Guha, N. Anita. *MuADDIBBS: A Hierarchical Outdoor Swarm System with Aerial Supervision Guiding Local Interaction*. 15th International Conference on Mechanical Engineering, BUET, 2025. SSRN

TOOLS

OpenFOAM and gmsh for simulation, previously ANSYS Fluent; Python (NumPy, SciPy, scikit-learn, PyTorch) for analysis; MATLAB/Simulink; C/C++ ; Linux, git, $\text{\LaTeX}$.